

Alkanes, Alkenes, and Alkynes

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1 Alkenes

Core Concept 1.1: Alkene Orbital Structure

The orbital structure of an alkene double bond is a combination of a σ bond and a π bond. Compared to the alkane single bond length of roughly 1.5 Å, the alkene double bond is shorter at approximately 1.3 Å.

<figure of sigma and pi bond side by side>

Core Concept 1.2: Alkene Bond Strength

The π bond of an alkene is relatively weaker than its partner σ bond because its corresponding antibonding π^* orbital sits closer to the bonding π orbital than the antibonding σ^* orbital with the σ bonding orbital. As a result, breaking an alkene π bond takes roughly 65 kcal mol^{-1} whereas the σ bond $108 \text{ kcal mol}^{-1}$.

<MO diagram figure>

When compared with alkanes, the typical alkene C-C bond strength is roughly 88 kcal mol^{-1} . Notice that the alkene σ bond is stronger than the alkane σ bond because the alkene σ bond uses sp^2 hybridized orbitals compared to the sp^3 orbitals of alkanes, as shown in FIGURE REFERENCE. This higher s orbital characteristic—specifically 33% vs 25%, respectively—means greater bond strength as e^- s are pulled in closer relative to p orbitals.

<figure>

Thus in the context of the pK_a values of substituent hydrogens of alkenes and alkanes, the alkenes show lower values compared to that of alkanes.

<figure of ethene and ethane with pK_a >

Therefore, in principle, the following reaction between methyllithium and an alkene will reach a product favoring equilibrium because the alkene is more acidic (*able to release a proton better*) than the alkane.

<figure of the reaction above>

However, in practice, the reaction in FIGURE REFERENCE will face regio- and stereoselectivity issues because the other substituent hydrogens can equally or more favorably dissociate compared to the one shown. In practice, chemists utilize the formation of **organometallics** (Core Concept 2.3) to overcome these problems.

<figure of the organometallics>

Core Concept 1.3: Relative Stability of Alkenes

Consider the **heat of hydrogenation** ($\Delta H_{\text{hydrogenation}}$) of following butene isomers to butane.

<figure of the butene to butane>

The relative greater stability of internal over terminal double bonds and trans over cis substituents arise from two reasons.

(a) **Hyperconjugation** allows the temporary charge imbalance arising from the transitionary state to be better distributed by neighboring molecular orbitals in the internal double bond butene compared to the terminal counterpart.

<figure>

(b) **Steric hindrance** between the cis methyl groups increase the relative instability of the butene compared to the trans counterpart.

<figure>

Thus the general order of alkene stability acts as follows.

<figure alkene stability order>

1.1 Alkene Synthesis

Core Concept 1.4: Elimination & Saytzev Rule

Synthesizing alkenes from alkanes through E2 elimination requires the consideration of several factors.

(a) Base Size

Consider an E2 reaction **with a small base** on the following RX alkane.

<figure with small base>

Notice the favoring of the internal double bond alkene compared to the terminal version because the advantages of hyperconjugation win over the minimal effects of steric hinderance due ot the use of a small base. In contrast, an E2 reaction **using large base** (*BELOW FIGURE REFERENCE*) favors the terminal version because steric hinderance exerts a comparatively larger influence on transition state stability enough to override to hyperconjugation favorability of the internal double bond product.

<figure with large base>

This phenomenon is termed the **Saytzev Rule**, which states that non-bulky bases lead to more stable internal double bonds. The corresponding energy diagrams of the two reactions are shown below to visually depict the energy levels of the reaction pariticipants throughotu the reaction.

<figure of energy diagrams>

(b) Stereoselectivity

The presence of **multiple stereoisomers**—specifically among the formed internal double bond alkenes when utilizing E2 reactions with small bases—is something a chemist must also consider. Although the E2 reaction is stereoselective, favoring the more stable trans isomers over the sterically hindered cis isomers, the relationship is not absolute (*some vis isomers do form*).

<figure>

(c) Other Factors

Other factors such as preferable precursor confirmations, namely rotamers, play a role as well.

<figure>

Keep in mind that elimination is also **stereospecific** as shown in the following figure.

<figure>

Core Concept 1.5: Alkenes from ROH By Dehydration

Alkenes can also be synthesized from exposing alcohols to acids and subsequently removing the resulting H₂O byproducts through dehydration.

<figure>

The mechanism behind the reaction—using sulfuric acid and ethanol as an example—is as follows.

<figure mechanism>

Due to the effects of steric hinderance and hyperconjugation, the usage of E1 or E2 and resulting stereoisomer product ratio differs across R_{primary} , $R_{\text{secondary}}$, and R_{tertiary} .

<figure>

A huge limitation to this is the formation of a **carbocation transition state**, which opens the door to **unintended rearrangement pathways** such as **hydride** (*Core Concept 3.1*) or **alkyl shifts** (*Core Concept 3.3*) if possible. As a result, this alkene synthesis pathway has limited utility in practice.

<figure>

1.2 Alkene Addition Reactions

Core Concept 1.6: Alkene Addition Reactions

Alkenes are the fundamental building blocks for organic synthesis and the preparation of polymer materials. Their defining feature is the **unsaturated π bond**, which reactions through **addition reactions**, wherein the π bond is broken and a diatomic or compound molecule is added across the relevant carbon atoms of the former alkene.

<figure of addition template>

Generally speaking, alkene addition reactions are $\Delta H^\circ < 0$, however since the nature of the reaction is a combination reaction wherein two reactant molecules form a single product molecule, $\Delta S^\circ < 0$, making it entropically unfavorable. In most reactions, the enthalpic favorability overrides the magnitude of entropic unfavorability. However, in some reactions or at high temperatures, the ΔG° turns positive, making these addition reactions thermodynamically unfavorable.

<figure of the ethene addition reaction>

1.2.1 Catalytic Hydrogenation

Core Concept 1.7: Catalytic Hydrogenation

Catalytic hydrogenation refers to the addition of H_2 to an alkene, effectively removing the alkene double bond and converting the target molecule into an alkane if the double bond in question was the only one. This process alone would require enormous amounts of pressure and extreme temperatures to occur, however **catalysts**—such as Pd/C or PtO_2 —enable a mechanism with lower E_a that chemists frequently use.

<figure>

It's also important to note that the hydrogens are **stereospecifically** added to the **same side (syn)** of the target double bond. As a result, the outputed product mixutre will be **racemic**.

<figure of the equation>

1.2.2 Electrophilic Additions

Core Concept 1.8: Eletrophilic Additions

Because the π bond is **e^- rich**, it readily **reacts with a polar reagent $A^{\delta+}-B^{\delta-}$** , where the partially positive end $A^{\delta+}$ is drawn to the e^- rich double bond.

In the scenario where $A^{\delta+}$ is a H^+ , the reaction that ensues is simply the reverse of an E1.

<figure> in the caption, indicate the anti and syn stereoisomers

Core Concept 1.9: Hydrohalogenation

Hydrohalogenation is a type of electrophilic addition wherein a **HX is added** to the alkene double bond where X is a halogen.

<figure>

Notice that the H^+ adds to the **less substituted** carbon because it results in a more substituted carbocation in the other carbon, favoring tertiary > secondary > primary carbocation formation. This phenomenon is called **Markovnikov's Rule** and the following energy diagram for the different stereospecific pathways depicts the stability of the different transition states nicely.

<figure of energy diagram>

Core Concept 1.10: Markovnikov Hydration

Markovnikov hydration refers to the conversion of an alkene to an alcohol through the **addition of H_2O** in the presence of an acid catalyst—the reverse of the alkene synthesis reaction covered in Core Concept 1.5.

<figure>

The underlying mechanism to the reaction is shown in the following. Notice how the H^+ is regenerated by the end of the reaction.

<figure of mechanism>

There are, however, bound to be a mixture of alkenes, transient intermediaries, and alcohols because Markovnikov hydration is at dynamic equilibrium with the pathway previously covered in Core Concept 1.5. Because the temporary presence of a carbocation introduces hydride shifts, catalytic H^+ will turn a single alkene species into a slurry of stereoisomeric alkenes, with ratios determined by relative thermodynamic stabilities primarily based on steric hinderance and hyperconjugation.

<figure of the alkene stereoisomers>

The full mechanism of equilibrium between an alcohol and its corresponding alkene products is shown in the following chemical equation using butanol/butene as an example.

<equilibrium equation> caption includes: leads to the rapid equilibrium of three alkene products

Core Concept 1.11: Halogenation

Halogenation is the addition of a diatomic halogen molecule X_2 to an alkene double bond. In contrast to the addition of HX (*Core Concept 1.9*)—which is inherently polarized due to the electronegativity differences— X_2 is not polarized inherently, but rather **gets polarized during its approach to the e^- rich alkene double bond**.

<outline chemical equation figure>

The underlying mechanism is shown in the following figure.

<mechanism>

Notice the formation of a halonium ring because the alternative carbocation state will want to distribute the burden of its positive charge. Because the second attack to the bridged halonium exclusively occurs to the backside of the first halogen attack, halogenation is **stereospecifically anti (not syn)**. Likewise, if the

second attack has thermodynamically equal possibility of attack either backside carbon—meaning that the initial alkene was perfectly **symmetrical**—the ensuing products will be an **racemic mixture**.

If the initial alkene is **asymmetrical**, the halonium ion intermediate is actually highly distorted, wherein the bond between the halogen and more substituted carbon is **longer**. Likewise, as the halogen pulls more e^- density towards itself the more substituted carbon is able to possess a greater δ^+ due to hyperconjugation. As a result, the halonium ring transition state **already has a high degree of carbocation** behavior and thus mimics more of a S_N1 attack characteristic although the backside attack is S_N2 by definition because of its exclusively anti result. Consequently, the backside nucleophilic attack occurs at the more substituted carbon, an interaction wherein the impact of charge is greater than steric hinderance on regioselectivity.

<figure of HOH attack>

On a different note, the intermediary halonium ring ions can also be intercepted by other nucleophiles such as HOH (FIG REFERENCE ABOVE) or ROH (FIG REFERENCE BELOW).

<figure of ROH attack>

Core Concept 1.12: Oxymercuration–Demercuration

Oxymercuration–demercuration is effectively the same as the **Markovnikov hydration** (Core Concept 1.10) process wherein a H_2O is added to remove an alkene double bond. However, oxymercuration–demercuration **avoids carbocation** formation, which prevents the unintended rearranged product alcohols and produces a purer product mix.

The process utilizes mercury (II) acetate and sodium borohydride in a two step chemical process to achieve its H_2O addition.

<figure of simple H2O equation>

(1) Oxymercuration

Similar to the **alkene halogenation** (Core Concept 1.11) process, the large mercury atom forms a bridged, unevenly distributed mercurinium ion rather than a full carbocation, forcing the subsequent H_2O attack to strictly occur at the more substituted carbon.

(2) Demercuration

Afterwards, the $NaBH_4$ acts as a reducing agent that cleaves the C–Hg bond and replaces it with a simple hydrogen atom.

Likewise, the usage of ROH allows the addition of a –OR group just as how a –OH was added by H_2O .

<figure with ROH>

When compared to Markovnikov hydration (Core Concept 1.10), oxymercuration–demercuration is the preferred method used by chemists in order to convert an alkene to an alcohol due to the purity of the resulting product mix.

<figure>

1.2.3 Hydroboration-Oxidations

Core Concept 1.13: Hydroboration-Oxidations

Hydroboration-oxidation is a method chemists use to **add H_2O** to a double bond of an alkene at the **less substituted carbon of the double bond** (remember that the Markovnikov hydration and oxymercuration-demercuration H_2O additions occur at the more substituted carbon). This process utilizes BH_3 to regioselectively bond to 3 alkenes and subsequently oxidate and cleave the resulting BR_3 to produce 3 ROH.

<simple equation figure>

(1) Hydroboration

When mixed with BH_3 , three alkenes bond to BH_3 with the **less substituted carbon paired to the boron**, occurring due to two major factors.

<figure>

• **Transition State Stabilization**

When the e^- rich π bond of the alkene reaches out to attack the empty unhybridized p orbital of the boron, the two carbon atoms of the double bond do not share the burden equally. The carbon that is **not** forming the C-B bond begins to lose its e^- density, creating a partial positive charge δ^+ during its transition state. Consequently, if the less substituted carbon forms the C-B bond, the more substituted carbon can better stabilize this charge through hyperconjugation, allowing for a more stable transition state relative to the alternative C-B bond formation.

<figure>

• **Bond Polarization**

Another subtle factor that plays a role is the inherent electronegativity difference between boron (2.0) and hydrogen (2.1), creating a polarized bond where boron is slightly positive (δ^+) and hydrogen is slightly negative (δ^-). As a result, the slightly positive boron more favorably pairs with the e^- rich attack from the less substituted alkene carbon and the slightly negative hydrogen more favorably pairs with the subsequently partial positive non-boron bonding alkene carbon.

<figure>

Another important characteristic of hydrobromination is the **syn** stereospecificity of the resulting alkylborane. The B and H that are attacked by an alkene π orbital are physically attached to one another and added to the double bond simultaneously, forcing them to attach to the same face of the alkene.

<figure>

(2) Oxidation

The resulting alkylborane R_3B is then oxidized by a solution of H_2O_2 and ^-OH which yields trialkyl borate $(\text{RO})_3\text{B}$ through the following alkyl shift mechanism.

<figure>

Finally, H_2O hydrolyzes the $(\text{RO})_3\text{B}$ with the help of a base to form the intended 3 ROH molecules with a borate ion as a byproduct.

<figure>

Often this entire process is called **anti-Markovnikov hydration** because it adds a H_2O molecule to the alkene but does so in a manner where the $-\text{OH}$ group ends up on the less substituted π bond carbon rather than the more substituted one.

1.2.4 Electrophilic Carbene Additions

Core Concept 1.14: Carbene Synthesis

Carbenes (:CR_2) are sp^2 hybridized 6e^- species with a lone pair and empty p orbital—the parent of which is methylene (:CH_2). They can be thought of as being derived by **deprotonating carbocations**.

<figure of deprotonation of a carbocation>

Because they are e^- deficient, carbenes are extremely unstable, short lived intermediates that act as powerful **electrophiles**. The following are the three primary pathways chemists use to generate these intermediates in the laboratory.

(a) **Methylene from diazomethane (CH_2N_2)**

<figure>

Mind that this reaction is practically irreversible because the resulting N_2 is incredibly stable, driving the reaction forward to cleanly leave behind highly reactive methylene.

(b) **Dichlorocarbene from chloroform**

<figure>

The above reaction utilizes α -elimination (*distinct from the standard β -eliminations E1 or E2 normally used to make alkenes*) through the strong base $(\text{CH}_3)_3\text{CO}^-$ and the departure of a highly electronegative chlorine atom.

(c) **Simmsons-Smith reagent**

As mentioned previously, free methylene is so wildly reactive that it can be incredibly dangerous to work with. The Simmsons-Smith reaction provides a much safer, controlled workaround.

<figure>

The treatment of diiodomethane with a zinc-copper couple forms an organometallic species that behaves like a hybrid between a stable molecule and a methylene like state.

Core Concept 1.15: Electrophilic Carbene Additions

Being the naturally reactive species that they are, carbenes can be added to alkenes in a **syn stereospecific manner** through the following general mechanism, forming carbon ring as a product.

<figure>

In practice, since carbene species are transient, this carbene addition must occur alongside the carbene synthesis processes (*Core Concept 1.14*).

(a) **Methylene from diazomethane (CH_2N_2)**

<figure>

(b) **Dichlorocarbene from chloroform**

<figure>

(c) **Simmsons-Smith reagent**

<figure>

1.2.5 Electrophile Oxidation

Core Concept 1.16: Peroxycarboxylic Acids

Peroxycarboxylic acids are a species of acids with the following general structure.

<figure>

When alkenes are treated with these compounds, the reaction results in a **syn addition of an oxygen** that forms a cyclic ether product as outlined below.

<figure of outline with red and blue coloring to represent charges and dashes and wedges to represent the stereospecificity>

The underlying **concerted** mechanism looks like the following.

<figure of mechanism>

It's also important to note that the **rates of oxacyclopropanation increases with alkyl substitution** because increased hyperconjugation allows to a greater degree of e^- donation from the electrophilic alkene π orbital.

<figure>

In practice, peroxycarboxylic acid addition usually serves as a preparatory step for subsequent modifications. Two common applications are shown below.

(a) **Double Nucleophile Syn-Addition**

<figure>

(b) **Anti Stereospecific Dihydroxylation**

<figure>

Core Concept 1.17: OsO₄ Syn-Dihydroxylation

The treatment of alkenes with OsO₄ and a subsequent reduction through an agent like H₂S results in syn-dihydroxylation along an alkene double bond.

<figure>

The underlying mechanism proceeds as follows, creating osmate ester intermediary product before the H₂S step reduction.

<figure>

The consequently used Os compound can be reoxidized by adding H₂O₂, allowing the Os compound to be **catalytic** (*not consumed by the reaction until H₂O₂ depletes*). This is helpful because Os is expensive

<figure of the example at the end>

Core Concept 1.18: Ozonolysis

Ozonolysis utilizes ozone (O₃)—a highly unstable and rapidly decaying species—and a reducing agent to perform a complete oxidative cleavage of an alkene double bond.

<figure>

Because ozone is such a transient molecular specie, it isn't available enough naturally chemists to work with it. To prevent this shortage, chemists put O₂ through an **arc discharge**—continuous, highly intense electrical

breakdown of a gas—to generate fresh stream of 2–4% O₃ gas safely in a lab.

<figure>

The following are three of the most common reducing agents used by chemists.

(a) **Zinc and Acid (Zn / CH₃COOH)**

<figure>

(b) **Catalytic Hydrogenation (H₂ / Pt)**

<figure>

(c) **Dimethyl Sulfide (DMS; (CH₃)₂S)**

<figure>

1.2.6 Radical Hydrobromination

Core Concept 1.19: Radical Hydrobromination

Radical hydrobromination refers to the **addition of an HBr** molecule to an alkene double bond with the assistance of RO – OR in an anti-Markovnikov manner, meaning that the Br is attached to the **less substituted carbon**. Recall that regular **hydrohalogenation** (*Core Concept 1.9*) follows Markovnikov regiochemistry where the halogen is attached to the more substituted carbon.

<figure of simple equation>

The underlying mechanism is shown in the following figure, started by peroxide radical initiators (*the O–O bond is weak at around 39 kcal mol^{–1}*)

(a) **Initiation Steps**

<figure>

(b) **Propagation Steps**

<figure> separate propagation I and II in the first

Notice that the Br* is added to the less substituted carbon, resulting in the resulting carbon radical to be better hyperconjugated than the alternative binding arrangement due to being more substituted.

This radical hydrobromination does not work for HCl or HI because one of the propagation steps are **endothermic** due the unique electronegative properties of Cl and I. Hence the radical mechanism is too slow and the ionic mechanism in the Markovnikov manner (*attaching to more substituted carbon*) is favored. Br, on the other hand, sits at the thermodynamic sweet spot between Cl and I. RSH anti-markovnikov addition works in the same manner as Br, sitting in that same sweet spot.

2 Alkynes

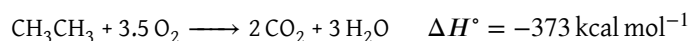
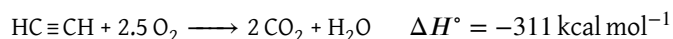
Core Concept 2.1: Alkyne Orbital Structure

An alkyne triple bond is composed of two perpendicular π bonds and a conjoined sp hybrid σ bond.

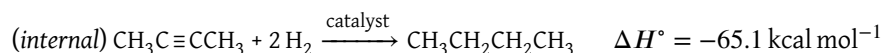
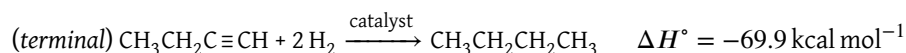
<figure>

Core Concept 2.2: Triple Bond Strength

Ethyne releases a massive $-311 \text{ kcal mol}^{-1}$ of heat when combusted. Compared to ethane (CH_3CH_3)—which releases -373 kcal/mol —ethyne releases a proportionally greater amount of energy on a per carbon basis because you don't have to spend extra energy to break the 4 additional C-H bonds in ethane.



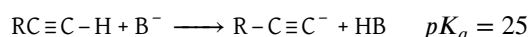
Due to the higher level of strain alkyne π bonds exist under, the full hydrogenation of the two alkyne π bonds releases more energy than that of two alkene π bonds (*which would be roughly $-60 \text{ kcal mol}^{-1}$*)



Notice the differences in net energy release due to the greater stability of the internal triple bond over the terminal version because of hyperconjugation.

Core Concept 2.3: Relative Acidity of Alkynes

In general hydrocarbons (alkanes and alkenes), the C-H bond is considered completely non-acidic. However, terminal alkyne C-H bonds defy this rule when treated with a sufficiently strong base, wherein the terminal H^+ can be plucked off.



<figure of the trend>

The above acidic trend is explained by the greater s orbital characteristics of the corresponding orbital hybridization. Since an s-orbital is spherical and sits much closer to the positively charged carbon nucleus than a directional, elongated p-orbital, the lone pair of electrons left behind after deprotonation is held very close to the positive nucleus. This electrostatic proximity stabilizes the negative charge immensely, making the corresponding alkyne conjugate base comparatively more stable.

Using this alkyne acidity, chemists are able to execute the following example applications

(a) Organolithium Reagents

<figure>

(b) Sodium Amide (NaNH_2)

<figure>

2.1 Alkyne Synthesis**Core Concept 2.4: Elimination E2 From Dihaloalkanes**

Performing E2 elimination twice on dihaloalkenes with adjacent halogen substituents allows for alkyne formation.

<figure>

Here is a specific example utilizing NaNH_2 as the base, which has to go through an extra aqueous work up step due the acidity of terminal alkynes (*Core Concept 2.3*) in the presence of sodium amide NaNH_2 .

<figure>

Core Concept 2.5: Alkylation of Alkynyl Metals

Terminal alkynes can be converted into powerful nucleophiles via deprotonation with strong bases, enabling C–C bond formation through substitution pathways that follow strict S_N2 rules.

<figure of the Li and I example>

The synthetic utility of alkynyl metals relies on R_{prim} halides or oxacyclopropanes (*epoxides*) as electrophilic substrates to minimize competing E2 elimination side-reactions. When utilizing unsymmetrical epoxides, the acetylide nucleophile regioselectively attacks the less sterically hindered carbon atom.

<figure of the last example reaction with the asymmetrical oxacyclopropane>

The following are various examples using this mechanism.

(a) **Carbonyl Addition with Grignard**

<figure>

(b) **Epoxide Opening with Lithium Acetylide**

<figure>

(c) **Double Grignard Addition of Acetaldehyde**

<figure>

2.2 Alkyne Reactions

2.2.1 Reductions

Core Concept 2.6: Complete Hydrogenation

The complete hydrogenation of an alkyne triple bond (*addition of $2H_2$*) progresses similarly alkene hydrogenation (*Core Concept 1.7*), utilizing a catalyst like Pt to help lower the E_a of the reaction. The resulting product is an alkane.

<reaction figure>

Core Concept 2.7: Partial Hydrogenation

Partial hydrogenation refers to the addition of only one H_2 wherein only one of the two π bonds are removed, resulting in either *cis* or *trans* product from a *syn* and *anti* addition, respectively.

- **Lindler's Catalyst (Syn Addition)**

<figure>

A *syn* addition and subsequent *cis* product requires the usage of a "poisoned" **Lindlar's catalyst**.

- **Sodium Reduction (Anti Addition)**

<figure>

The underlying mechanism for the above reduction through sodium is shown below.

(1) **e^- Transfer**

<figure> (2) First Protonation <figure> (3) Second e⁻ Transfer <figure> (4) Second Protonation <figure>
--

2.2.2 Electrophilic Additions (Markovnikov)

Core Concept 2.8: HX Anti Addition
Core Concept 2.9: X₂ Anti Addition
Core Concept 2.10: Mercury Catalyzed Markovnikov Hydration
Core Concept 2.11: Radical HBr Anti-Markovnikov Addition
Core Concept 2.12: Hydroboration-Oxidation